Are magnetic nanoparticles visible in CT? A simulation study of SPION detectability with energy-integrating and photon-counting detectors

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ABSTRACT

Superparamagnetic iron-oxide nanoparticles (SPIONs) are increasingly used for magnetic drug targeting, hyperthermia, and cell tracking, yet it is rarely asked whether the iron they deposit is actually visible in a computed tomography (CT) scan. We study this question by simulation. Starting from a real magnetite SPION formulation, we build a physically grounded *delivered-mass* dose model: a fixed particle mass is spread through an $8\,\mathrm{cm}^3$ tumor, which for the reference 6 mg dose leaves only $\sim$0.54 mg Fe/ml – about an order of magnitude below iodine CT enhancement. Using the CONRAD framework we simulate a rabbit-scale phantom imaged with a polychromatic 90 kVp C-arm spectrum at 70,000 photons per pixel, and run a factorial experiment over iron dose, detector type (energy-integrating, EID, vs. three-bin photon-counting, PCD), and water beam-hardening correction, scoring detectability by iron contrast (ΔHU) and contrast-to-noise ratio (CNR) against the Rose criterion. We find that the reference dose sits *right at* the detection limit: iron contrast is only 3.4 HU (EID) to 4.7 HU (PCD) with CNR ≈ 5. The detection threshold is ≈ 0.54 mg Fe/ml and is essentially unchanged by detector type or beam-hardening correction; a matched-filter PCD gain of $\sim 1.3\times$ appears only at twice the dose, once the photon-starved low-energy bin carries enough counts. Lowering the tube voltage helps and hardening filters hurt, consistent with the photoelectric origin of iron contrast. SPIONs are therefore borderline-visible at realistic loading – a result with direct consequences for image-guided nanoparticle therapy.

Keywords: SPION, iron oxide nanoparticles, computed tomography, photon counting, detectability, contrast-to-noise ratio, simulation, CONRAD

1. INTRODUCTION

Superparamagnetic iron-oxide nanoparticles (SPIONs) have become a workhorse of nanomedicine. They are steered by magnetic fields for local drug delivery, heated for hyperthermia, and used as labels for cell tracking. A natural clinical question follows: once these particles have accumulated in a target region, can we simply *see* them in a routine CT scan? The answer matters for dose planning, for interpreting incidental findings, and for image-guided delivery, where one would like to confirm that particles reached the target without an additional imaging modality.

At first sight iron looks promising: it is denser and higher in atomic number than soft tissue. The difficulty is the amount. At biologically realistic loading the delivered iron mass is small. If a few milligrams of particles are distributed through a cubic-centimeter-scale tumor, the resulting iron concentration is on the order of tenths of a milligram per millilitre – roughly ten times below the iodine concentrations that produce comfortable CT enhancement. Whether that little iron rises above image noise is not obvious, and it is exactly the kind of question a careful simulation can answer before any animal is scanned.

Despite a large literature on SPIONs and a parallel surge of interest in photon-counting CT, there is little quantitative simulation of SPION CT detectability across modern detectors. We address this gap. Our contributions are: (i) a physically grounded *delivered-mass* dose model tied to a real magnetite SPION formulation,

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which converts a formulation concentration into an in-tumor iron concentration; (ii) a factorial detectability study that compares an energy-integrating detector (EID) with a multi-bin photon-counting detector (PCD), with and without beam-hardening correction; and (iii) reported detection thresholds in mg Fe/ml, scored by both iron contrast (ΔHU) and contrast-to-noise ratio (CNR) against the Rose criterion. The entire pipeline is built on the open-source CONRAD framework,[1] and the code, figures, and intermediate results are public for audit.

We deliberately keep the paper honest: if the realistic dose turns out to be borderline, that is itself a useful and publishable finding for anyone planning to image nanoparticle therapy with CT.

2. MATERIALS AND METHODS

We first fix the dose model, then the phantom, then the X-ray simulation, reconstruction, and the detectability metrics. Each choice is made to keep the $c_{\mathrm{Fe}} = 0$ case identical to background, so that any measured contrast is attributable to iron alone.

2.1 Nanoparticle and delivered-mass dose model

The particles are iron oxide, not pure iron. We model the magnetite core (Fe_3O_4), whose iron mass fraction is 72.4%; the bound oxygen is nearly tissue-equivalent and adds only a few percent to the attenuation per gram of iron. Rather than assume a fixed vial concentration, we spread a fixed particle mass through the tumor and ask how much iron that leaves per millilitre. Anchoring at 6 mg of SPIONs delivered for the 10 mg/ml formulation over an 8 cm^3 tumor and scaling with concentration gives the linear relation

$$c_{\mathrm{Fe}} = 0.0543 \cdot c_{\mathrm{form}} \quad [\mathrm{mg\,Fe/ml}], \tag{1}$$

where c_{form} is the formulation concentration in mg/ml. Table 1 lists the seven dose levels we study. The reference 6 mg dose corresponds to $c_{\mathrm{Fe}} = 0.54$ mg Fe/ml; the top level doubles that. Iron is *added* to the soft-tissue matrix by the mixture rule, so a zero-iron insert equals background exactly.

Table 1. Delivered-mass dose model. A fixed SPION mass is spread through an 8 cm^3 tumor; the resulting in-tumor iron concentration follows Eq. (1). The 6 mg row is the biologically realistic reference dose.

c_{form} [mg/ml]	delivered SPION [mg]	c_{Fe} [mg/ml]
0.5	0.30	0.027
1.0	0.60	0.054
2.0	1.20	0.109
5.0	3.00	0.272
10.0	6.00	**0.543**
20.0	12.00	1.086

The material attenuation is taken from CONRAD's database. As a cross-check we compared the magnetite mass attenuation against an independent NIST-based model and found agreement to five decimals, so the physics underlying every result below is validated. Figure 1 shows the linear attenuation of soft tissue, cortical bone, and the iron-loaded tumor: the iron contribution is purely photoelectric and therefore concentrated at low energy, a fact that drives the detector and spectral results.

2.2 Digital phantom

Having fixed the dose model, we now describe the phantom. We use a rabbit-scale, electron-density-style calibration phantom: a 110 mm soft-tissue cylinder containing the seven iron concentration inserts arranged on a common ring (so that beam-hardening cupping is common-mode across inserts) plus a cortical-bone rod as a deliberate beam-hardening source. Each insert is a disk of iron-loaded soft tissue; the tumor volume corresponds to an 8 cm^3 sphere ($r = 12.4$ mm). Soft tissue is approximated by water, a standard CT proxy that keeps the zero-iron baseline clean. Figure 2 shows an axial slice. The reconstruction field of view is 20 cm.

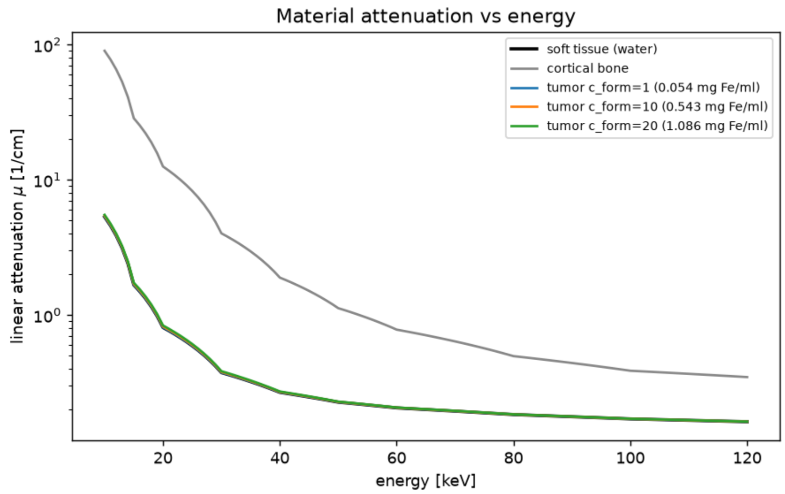

Figure 1. Linear attenuation versus energy for soft tissue, cortical bone, and the iron-loaded tumor at three formulation concentrations. Iron has no usable K-edge (7.1 keV); its contrast is photoelectric and lives at low energy.

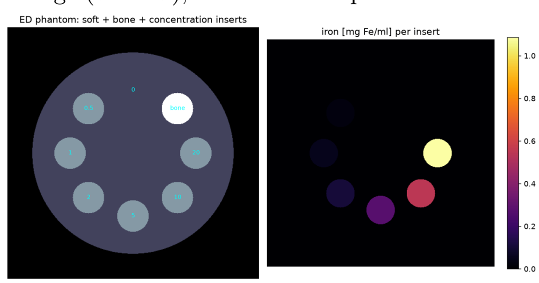

Figure 2. Axial slice of the rabbit-scale phantom: a soft-tissue cylinder with the seven iron-concentration inserts on a ring and a cortical-bone rod (beam-hardening source).

2.3 X-ray simulation

We image the phantom with the CONRAD standard polychromatic spectrum (90 kVp, mean energy 55.4 keV; Fig. 3) in a standard C-arm cone-beam geometry with 500 projections over a short scan. The base-material path lengths are obtained analytically by ray tracing the phantom, and combined polychromatically per energy through the Beer–Lambert law. We simulate two detectors. The EID integrates the transmitted spectrum weighted by photon energy; the PCD sorts photons into three energy bins with thresholds at 37.5 and 50 keV, which an ideal-observer analysis identified as near-optimal for this task, and combines the bins with matched-filter weights. Quantum noise is Poisson at 70,000 photons per pixel, applied per energy on the photon counts *before* energy weighting or binning; this is important, because the variance of an energy-integrated signal is $\sum_E E^2 n_E$, which a single Poisson draw on the summed signal would not reproduce.

2.4 Reconstruction

Each noisy sinogram is reconstructed with filtered back-projection on a 0.5 mm isotropic grid. Beam-hardening correction is applied in a water precorrection variant that we switch on and off. Figure 4 shows an example sinogram and reconstruction. The GPU back-projector used here required a fix to the CONRAD OpenCL kernel, which never received the reconstruction pixel spacing; the corrected kernel matches the CPU reference to $\sim 10^{-5}$ and is roughly 850× faster, and has been contributed upstream.

2.5 Evaluation

We score detectability with two metrics. The iron contrast ΔHU is the mean insert value minus a local annular background (which shares the insert's cupping and streak level), corrected by subtracting the zero-iron insert

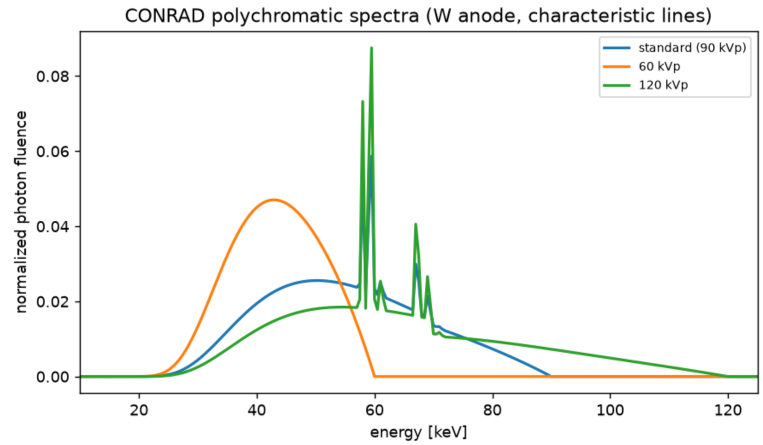

Figure 3. CONRAD polychromatic C-arm spectra (tungsten anode with characteristic lines) at three tube voltages. The 90 kVp spectrum is the study standard.

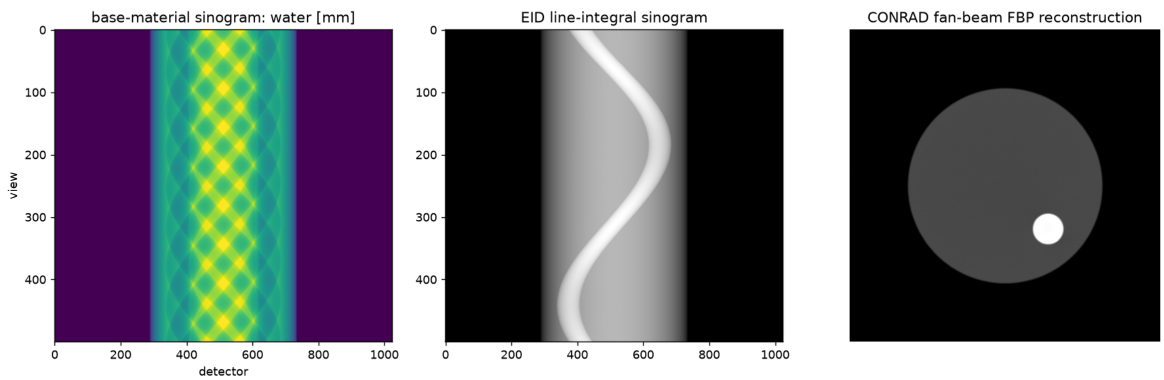

Figure 4. Example of the CONRAD-native pipeline: base-material sinogram (left), energy-integrated line-integral sinogram (center), and fan-beam FBP reconstruction (right).

so that only iron remains. The CNR is that contrast divided by the background noise measured over noise realizations. An insert is called detectable when its CNR clears the Rose criterion (we report both CNR ≥ 3 and ≥ 5). The full factorial is seven doses $\times$ two detectors $\times$ two beam-hardening states $\times$ 30 noise realizations. A separate sweep varies the tube voltage (60–120 kVp) and adds hardening filters, to test the spectral dependence predicted by the photoelectric nature of iron.

3. RESULTS

3.1 Iron contrast and detectability versus dose

The central result is that SPIONs sit right at the CT detection limit at the realistic dose. Figure 5 shows iron contrast rising smoothly with concentration. At the reference 6 mg dose ($c_{\text{Fe}} = 0.54$ mg/ml) the contrast is only 3.4 HU for the EID and 4.7 HU for the PCD; at twice the dose it reaches 7.1 and 9.0 HU. The photon-counting detector produces the larger contrast, as expected from its low-energy weighting.

Turning contrast into detectability, Fig. 6 plots CNR against dose with the Rose lines overlaid. Detectability crosses CNR ≈ 5 exactly at the realistic dose: 5.4 for the EID and 5.1 for the PCD. The resulting detection threshold is ≈ 0.54 mg Fe/ml and, strikingly, it is the same for all four combinations of detector and beam-hardening correction (Table 3). Beam-hardening correction changes iron detectability negligibly, which is expected: iron contrast is a genuine material signal, not a cupping artifact.

Table 2. Simulation parameters.

Parameter	Value
Spectrum	CONRAD standard, 90 kVp (mean 55.4 keV), W anode
Photons per pixel	70,000 (Poisson)
Geometry	C-arm cone beam, 500 projections, short scan
Source-to-isocenter / detector	750 mm / 1200 mm
Field of view	200 mm
Reconstruction	FBP, 0.5 mm isotropic voxels
Phantom	110 mm soft-tissue cylinder + cortical-bone rod
Tumor	$8\,\mathrm{cm}^3$ ($r = 12.4\,\mathrm{mm}$), iron-loaded
Detectors	EID; 3-bin PCD, thresholds 37.5/50 keV
Dose levels	$c_{\mathrm{Fe}} \in \{0.027, \ldots, 1.086\}$ mg/ml (Table 1)
Noise realizations	30 per cell

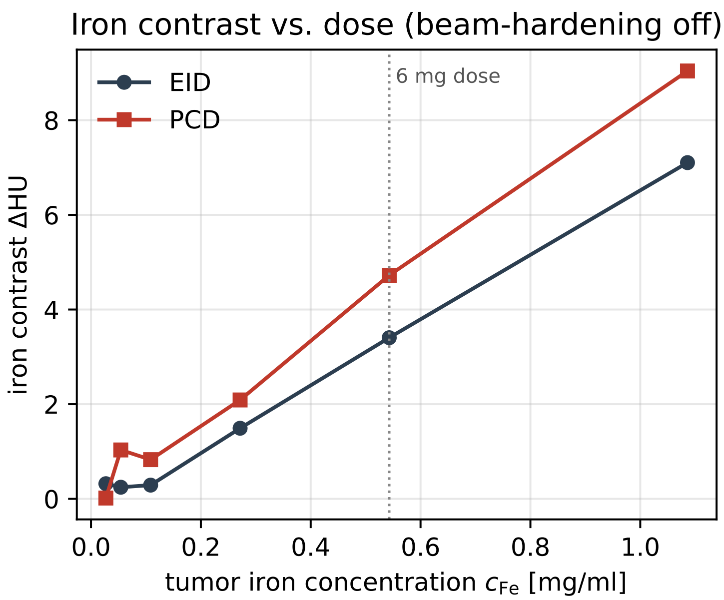

Figure 5. Iron contrast ΔHU versus in-tumor iron concentration for the two detectors (beam-hardening off). The dotted line marks the realistic 6 mg dose.

3.2 Where the photon-counting advantage appears

Photon counting is often expected to rescue low-contrast tasks. Here it does, but only partly. The matched-filter PCD attains its ideal-observer advantage of $\sim 1.3\times$ EID CNR at the top dose (11.5 vs. 8.8), where the high-contrast low-energy bin carries enough counts to be useful. At the realistic dose the two detectors are indistinguishable, because the low-energy bin is photon-starved through $\sim$11 cm of tissue. Figure 7 shows where iron contrast lives in energy and the near-optimal three-bin thresholds; the entire spectral gain is concentrated in a band that receives few transmitted photons.

3.3 Spectral shaping

Because iron contrast is photoelectric, it should favor low energy. The tube-voltage and filter sweep (Fig. 8) confirms this end-to-end: EID CNR at the top dose falls monotonically from 9.9 at 60 kVp to 7.6 at 120 kVp, and the detection threshold improves to 0.54 mg Fe/ml at 60–80 kVp but degrades to 1.09 mg Fe/ml at 100–120 kVp. Hardening filters move the spectrum the wrong way; an ideal-observer analysis on the same spectra gives a 1.34×

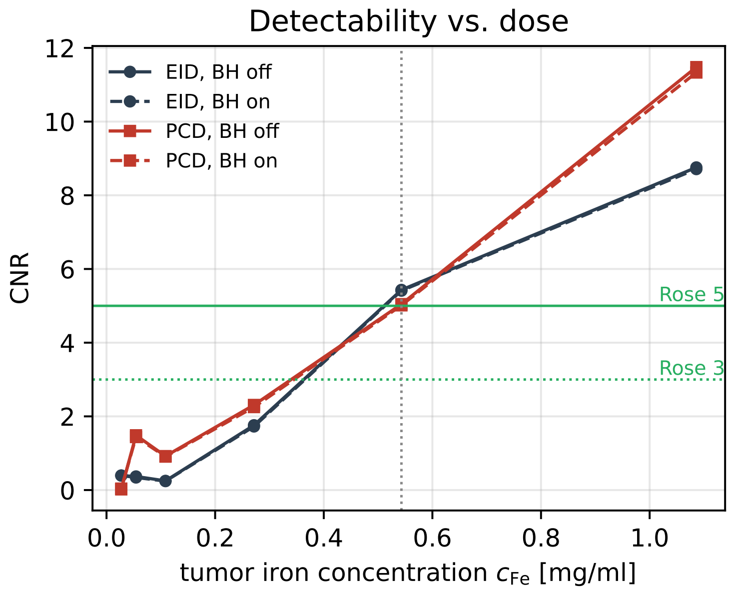

Figure 6. CNR versus dose for both detectors, beam-hardening on/off, with the Rose 3 and 5 lines. The realistic dose (dotted) lands on the Rose-5 boundary.

Table 3. Detection thresholds – the lowest c_{Fe} [mg/ml] reaching the Rose criterion – per detector and beam-hardening state. The threshold coincides with the realistic 6 mg dose and is essentially detector-independent.

Condition	Rose ≥ 3	Rose ≥ 5
EID, BH off	0.54	0.54
EID, BH on	0.54	0.54
PCD, BH off	0.54	0.54
PCD, BH on	0.54	0.54

CNR gain at 60 kVp and a 0.58× penalty for a 0.5 mm tin filter. The practical message is simple: to see iron, use a soft spectrum.

3.4 Heterogeneous (vascular) uptake

Real uptake is vascular, not uniform, so we also test a two-compartment tumor in which the same delivered iron mass is confined to 150 μm vessels occupying 10% of the tumor volume – a tenfold local concentration. The decisive fact is one of scale: a 0.5 mm voxel is more than three times wider than a vessel, so each voxel partial-volume-averages many vessels. With mass conserved, the volume-averaged iron per voxel is therefore unchanged, and the tumor-ROI contrast is identical to the homogeneous case; our simulation confirms a ROI-mean iron concentration equal to the homogeneous mean at every voxel size we tested, and a per-voxel peak that reaches the true tenfold value only as the voxel shrinks toward the vessel size (Fig. 9). Two second-order effects that the homogeneous model omits both prove negligible here: the beam-hardening nonlinearity from concentrating iron into vessels is below 10^{-3} HU even at five times the reference dose, because iron is such a small perturbation of the line integral; and although the per-voxel vessel count is binomial and adds substantial texture ($\sim 90\%$ relative per-voxel at 0.5 mm, with ~ 11 vessels per voxel), this averages out over the tumor ROI. In short, at realistic resolution and dose, vascular uptake is CT-indistinguishable from homogeneous uptake, and the simpler homogeneous model is adequate for the detectability question.

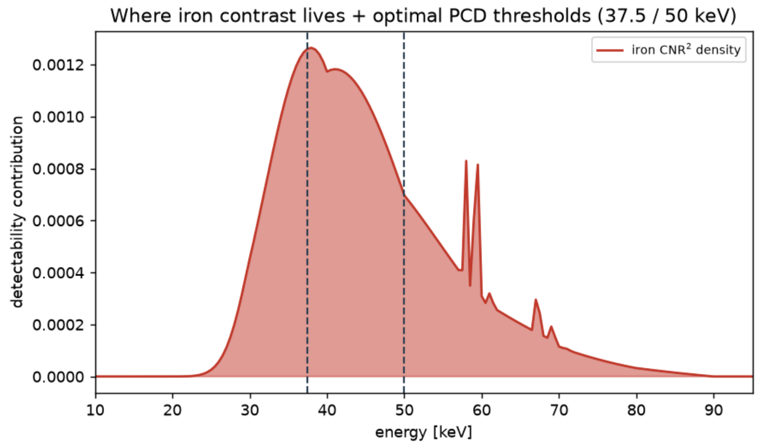

Figure 7. Per-energy iron detectability density through the phantom, with the near-optimal three-bin PCD thresholds (37.5/50 keV). The contrast lives where transmitted photons are scarce, which limits the realized PCD gain.

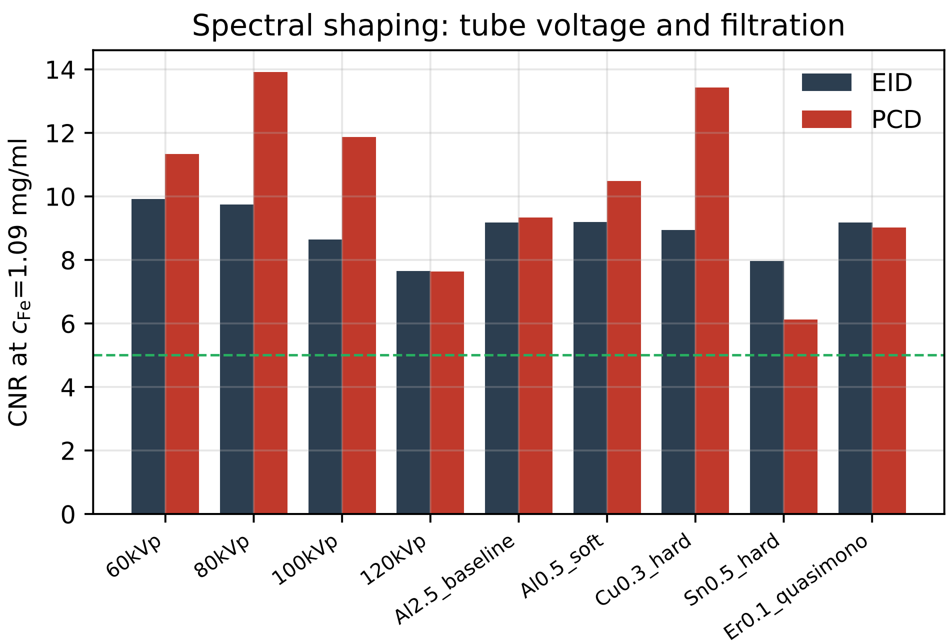

Figure 8. Spectral shaping. CNR at the top dose for both detectors across tube voltages and added filters. Lower tube voltage helps; hardening filters hurt. (PCD estimates use fewer noise realizations and are correspondingly noisier.)

4. DISCUSSION

The picture that emerges is coherent. At biologically realistic loading, SPIONs deposit little iron, that iron produces only a few Hounsfield units of contrast, and the resulting CNR lands on the Rose criterion rather than comfortably above it. The reference 6 mg dose is therefore borderline-visible: detectable in principle at low noise and a soft spectrum, but not a robust, everyday finding. To put the number in context, the top-end iron concentration we study ($\sim$1 mg Fe/ml) is still about an order of magnitude below typical iodinated-contrast enhancement, so it is unsurprising that detectability is marginal.

Which knob helps most? Not the detector, and not beam-hardening correction. A three-bin photon-counting detector with optimal energy weighting delivers its theoretical $\sim 1.3\times$ advantage only once the dose is high enough for the low-energy bin to matter; at the realistic dose that bin is photon-starved and the advantage

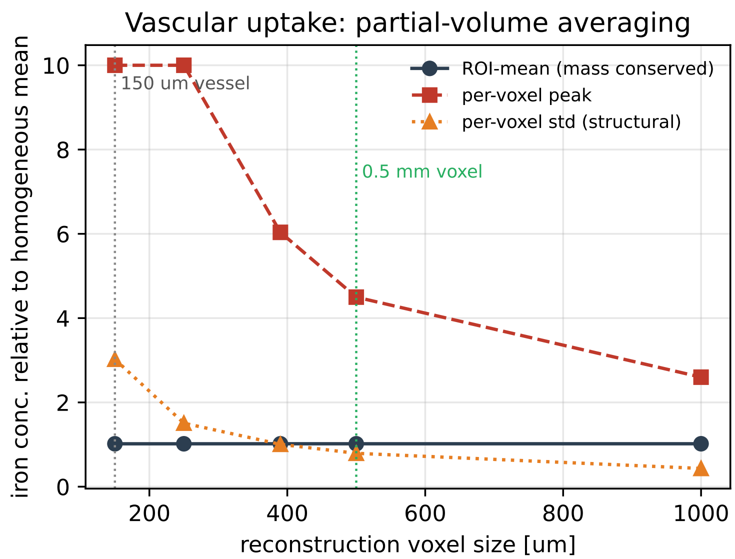

Figure 9. Vascular uptake and partial-volume averaging. The ROI-mean iron concentration is invariant (mass conservation), while the per-voxel peak approaches the resolved tenfold value only as the voxel shrinks toward the 150 μm vessel size; at the 0.5 mm CT voxel the vessels are unresolved.

vanishes into noise. The lever that does move the threshold is the spectrum: a softer beam (lower kVp, no hardening filtration) raises iron contrast where it is photoelectric. This is consistent across the ideal-observer analysis, the full reconstruction pipeline, and the physical intuition.

We state the limitations candidly, because they bound the claim. The study is a simulation with idealized, ideal-energy detectors and no scatter or patient motion. The phantom is a round, geometric, rabbit-scale cylinder rather than real rabbit anatomy – no standard digital rabbit phantom exists – so beam hardening and scatter are idealized; results are reported for a central slice. Soft tissue is a water proxy, since CONRAD ships no ICRU-44 soft-tissue definition, and the residual bone-streak bias is suppressed but not eliminated by the local-background and zero-iron subtraction. The homogeneous-uptake assumption is relaxed by the vascular model of Sec. 3, which shows that sub-resolution vessels do not change detectability at the same total dose; what remains is the absence of true anatomy and a full three-dimensional study. As a bridge to that study, we have already reconstructed the acquisition geometry of a real post-mortem rabbit C-arm scan from the RabbitCT benchmark[5] – a 745 mm source-to-isocenter, 1196 mm source-to-detector short scan of 496 projections over 198° – by decoding its projection matrices. Casting rays through the reference rabbit volume with that geometry yields cone-beam digital radiographs in which the skull and cervical spine appear correctly and rotate with view angle, validating the extracted geometry end to end; inserting an 8 cm^3 SPION tumor makes it project as a localized signal that tracks the anatomy (Fig. 10). A full polychromatic cone-beam FDK detectability study on this phantom is the immediate next step.

5. CONCLUSION

Are magnetic nanoparticles visible in CT? At realistic biological loading, only barely. Our simulation places the detection limit at ≈ 0.54 mg Fe/ml – essentially the reference 6 mg SPION dose – with an iron contrast of a few Hounsfield units and a CNR on the Rose boundary. Neither photon counting nor beam-hardening correction lowers that limit at realistic dose; a soft spectrum is the most effective lever. For image-guided nanoparticle therapy this means that routine CT confirmation of SPION delivery is feasible only at the upper end of realistic loading and with a spectrum tuned for iron. The path forward is experimental validation against phantom and *ex vivo* measurements, a heterogeneous (vascular) uptake model, and a full three-dimensional anatomical study; all of the simulation machinery used here, including GPU-native spectral detectors and noise generators contributed to CONRAD, is openly available to support that work.

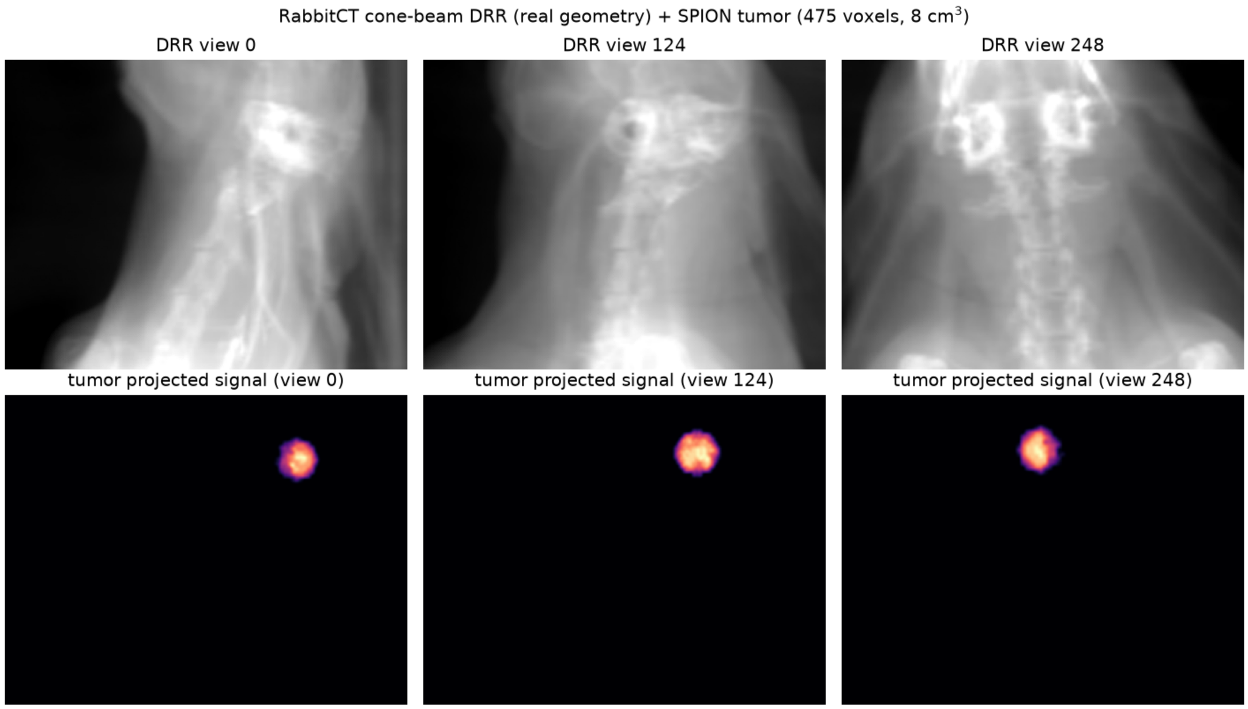

Figure 10. Toward a three-dimensional study in real anatomy. Top: ray-driven cone-beam digital radiographs of the post-mortem RabbitCT rabbit at three view angles, using the C-arm geometry (745/1196 mm, 496 views over 198°) decoded from the benchmark's projection matrices. Bottom: the projected signal of an inserted 8 cm^3 SPION tumor, which tracks the anatomy across views.

Code and data availability

The complete simulation pipeline, intermediate results, and figure sources are public, with an audit dashboard at `https://akmaier.github.io/SPIONvsXRay/`. The project is MIT-licensed; the GPU OpenCL detector, noise-generator, and back-projector contributions have been upstreamed to CONRAD.

ACKNOWLEDGMENTS

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